

D2 Sequences A level only

Name: _____

Class: _____

Date: _____

Time: **51 minutes**

Marks: **43 marks**

Comments:

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Q1.

A periodic sequence is defined by $U_n = \sin\left(\frac{n\pi}{2}\right)$

State the period of this sequence.

Circle your answer.

8

2π

4

π

(Total 1 mark)

Q2.

The n th term of a sequence is u_n . The sequence is defined by

$$u_{n+1} = pu_n + q$$

where p and q are constants.

The first two terms of the sequence are given by $u_1 = 96$ and $u_2 = 72$.

The limit of u_n as n tends to infinity is 24.

(a) Show that $p = \frac{2}{3}$.

(4)

(b) Find the value of u_3 .

(2)

(Total 6 marks)

Q3.

The n th term of a sequence is u_n . The sequence is defined by

$$u_{n+1} = pu_n + q$$

where p and q are constants.

The first two terms of the sequence are given by $u_1 = 60$ and $u_2 = 48$.

The limit of u_n as n tends to infinity is 12.

(a) Show that $p = \frac{3}{4}$ and find the value of q .

(5)

(b) Find the value of u_3 .

(1)

(Total 6 marks)

Q4.

The n th term of a sequence is u_n .

The sequence is defined by

$$u_{n+1} = 6 + \frac{2}{5}u_n$$

The first term of the sequence is given by $u_1 = 2$.

(a) Find the value of u_2 and the value of u_3 .

(2)

(b) The limit of u_n as n tends to infinity is L .

Write down an equation for L and hence find the value of L .

(3)

(Total 5 marks)

Q5.

The n th term of a sequence is u_n .

The sequence is defined by

$$u_{n+1} = ku_n + 12$$

where k is a constant.

The first two terms of the sequence are given by

$$u_1 = 16 \quad u_2 = 24$$

(a) Show that $k = 0.75$.

(2)

(b) Find the value of u_3 and the value of u_4 .

(2)

(c) The limit of u_n as n tends to infinity is L .

(i) Write down an equation for L .

(1)

(ii) Hence find the value of L .

(2)

(Total 7 marks)

Q6.

The n th term of a sequence is u_n .

The sequence is defined by

$$u_{n+1} = pu_n + q$$

where p and q are constants.

The first three terms of the sequence are given by

$$u_1 = -8 \quad u_2 = 8 \quad u_3 = 4$$

- (a) Show that $q = 6$ and find the value of p .

(5)

- (b) Find the value of u_4 .

(1)

- (c) The limit of u_n as n tends to infinity is L .

- (i) Write down an equation for L .

(1)

- (ii) Hence find the value of L .

(2)

(Total 9 marks)

Q7.

The n th term of a sequence is u_n .

The sequence is defined by

$$u_{n+1} = pu_n + q$$

where p and q are constants.

The first three terms of the sequence are given by

$$u_1 = 200 \quad u_2 = 150 \quad u_3 = 120$$

- (a) Show that $p = 0.6$ and find the value of q .

(5)

- (b) Find the value of u_4 .

(1)

- (c) The limit of u_n as n tends to infinity is L . Write down an equation for L and hence find the value of L .

(3)

(Total 9 marks)



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